

Technology Stack

Date	5 July 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	2 Marks

Technology Stack Details :

Technology Stack

The **Flood Prediction System using Machine Learning** is developed using modern technologies for machine learning, web application development, data storage, and cloud deployment. The selected technology stack ensures efficient flood prediction, easy maintenance, and a user-friendly experience.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript, Bootstrap	Provides a responsive, user-friendly, and interactive interface for entering weather data and displaying flood predictions.

2	Backend / Server-Side	Python, Flask	Flask provides a lightweight framework for integrating machine learning models and handling prediction requests efficiently.
3	Database / Data Storage	CSV Dataset / SQLite	Stores historical weather and flood data required for training and testing machine learning models. SQLite is lightweight and easy to manage.
4	Cloud / Hosting / Deployment	IBM Cloud	Enables cloud deployment, scalability, and remote accessibility of the Flood Prediction System.

5	Version Control & CI/CD	GitHub	Supports version control, collaboration among team members, and project source code management.
6	Third-Party APIs / Other Tools	Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib	Used for data preprocessing, machine learning model training, evaluation, and data visualization to improve flood prediction accuracy.